Week 4, July-Sept 2017
MOOCS Decision Modeling Course

Waiting Lines (Queuing Theory)

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Queuing Problems (Continued)

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A Variation of the Example

In a service centre only one person can get service and another one can wait. Any other arriving customer has to wait outside as there is only one place available inside. The customers arrive randomly at 8 per hour. The service is exponential and takes 6 minutes on the average. Find the probability that an arriving customer will have to wait outside.

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A Variation of the Example

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Answer: Average arrival rate $\lambda = 8$ per hour.

Average service time $1/\mu = 6$ minutes, and hence $\mu = 10$/hour.

Hence, the utilization factor, $\rho = \lambda/\mu = 8/10 = 4/5$

So, the probability that an arrival will find the place free

$= p(0) + p(1) = (1 - \rho) + \rho\,(1 - \rho) = (1 - (4/5)) + (4/5)^* (1 - (4/5))$

$= (1/5) + (4/25) = 9/25$

An arriving customer has to wait outside when there is no place free inside. So, probability that an arriving customer has to wait outside:

= 1 – probability that the arrival finds the place free

$= 1 - (9/25) = 16/25 = 64\%$.

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A Problem on Uniform Distribution

In a suburban station, trains arrive regularly every 30 minutes beginning 6.30 AM in the morning. A man, who does not know the train timings, arrives randomly (uniformly distributed) between 7.40 AM and 8.15 AM every morning. What is the probability that the man has to wait for more than 10 minutes for a train?

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A Problem on Uniform Distribution

In a suburban station, trains arrive regularly every 30 minutes beginning 6.30 AM in the morning. A man, who does not know the train timings, arrives randomly (uniformly distributed) between 7.40 AM and 8.15 AM every morning. What is the probability that the man has to wait for more than 10 minutes for a train?

Answer:

The man has to wait for more than 10 minutes for a train only if the arrival time is between 7.40 AM and 7.50 AM or between 8.00 AM and 8.15 AM. This is because there is a train exactly at 8 AM.

If X is a random variable that denotes the number of minutes past 7.40 AM that the passenger arrives, the desired probability is:

$$P(0 < X < 10) + P(20 < X < 35)$$

Now X is a uniform random variable on (0, 30). Therefore, the desired probability is given by:

$$F(10) + F(35) - F(20) = 10/35 + 1 - 20/35 = 25/35 = 5/7$$

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Equivalence of Poisson & Exponential

Show that if the number of arrivals of customers to a facility follows the Poisson distribution, then the inter-arrival time of the customers to the facility follows the exponential distribution.

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Equivalence of Poisson & Exponential

Show that if the number of arrivals of customers to a facility follows the Poisson distribution, then the inter-arrival time of the customers to the facility follows the exponential distribution.

Answer

It is indeed true that if the number of arrival of customers to a facility follows the Poisson distribution, then the inter-arrival time of the customers must follow the exponential distribution.

Poisson Distribution:

Probability mass function $p(x) = \dfrac{(\lambda t)^x e^{-\lambda t}}{x!}$

Cumulative distribution function : $F(x) = \left(\sum_{n=0}^{x} \dfrac{(\lambda t)^n e^{-\lambda t}}{n!} \right)$

Exponential distribution:

Probability density function $f(x) = \lambda e^{-\lambda x}$

Cumulative distribution function P(x ≤ t) : $F(x) = 1 - e^{-\lambda x}$

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Equivalence of Poisson & Exponential

Show that if the number of arrivals of customers to a facility follows the Poisson distribution, then the inter-arrival time of the customers to the facility follows the exponential distribution.

Answer

Now let us assume customers arrive at 2 per hour, i.e. $\lambda = 2$/hour.

To find probability that there would be no arrivals in next one hour

From Poisson Distribution: $\lambda = 2$/hour; $t = 1$ hour, hence $\lambda t = 2$

Probability of no arrival in the next hour: $p(0) = (2^0 e^{-2})/0! = e^{-2} = 0.135 = 13.5\%$.

From Exponential Distribution, $\lambda = 2$/hour; $t = 1$ hour.

Probability of no arrival in the next hour = $P(X > 1)$

$= 1 - P(x \leq 1) = 1 - (1 - e^{-2*1}) = e^{-2} = 0.135 = 13.5\%$

We see that the same result is obtained from Poisson as well as Exponential Distribution.

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Memory-less Property

Lack of Memory $P(X > t + \Delta t \mid X > \Delta t) = P(X > t)$

$$LHS = e^{-\lambda(t+\Delta t)} / e^{-\lambda(\Delta t)} = e^{-\lambda t} = RHS$$

For **exponential distribution**, probability of waiting a given length of time is **not affected** by how long one has already waited!

If someone has **already waited Δt minutes**, then probability that the person waits **another t minutes** is the same as that of someone else who is going to **wait t minutes** from the beginning.

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Suppose in a parking lot on a Sunday, last arrival was at 12 noon. λ = 1/hour. Inter-arrival time is exponential. **Nobody has arrived till 1 PM.** What is probability that nobody arrives before 2 PM?

Method 1: $P(X > 2 \mid X > 1) = e^{-1*2} / e^{-1*1} = e^{-2} / e^{-1} = e^{-1} = 0.368$

Method 2: As nobody arrived till 1 PM, Using **memory-less** property of exponential distribution: $e^{-1} = 0.368$

Queuing Example

In a parking lot on a Sunday, last arrival was at 12 noon. Arrival rate λ = 1/hour; service rate μ = 2/hour. No. of arrivals is Poisson and Inter-arrival time is exponential. Service time is also exponential.

a) What is probability that next arrival is before 1 PM?

b) What is probability that next arrival is between 1 and 2 PM?

c) What is probability that next arrival after 2 PM?

d) Nobody arrived till 1 PM. Probability of next arrival before 2 PM?

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Queuing Example

In a parking lot on a Sunday, last arrival was at 12 noon. Arrival rate λ = 1/hour; service rate μ = 2/hour. No. of arrivals is Poisson and Inter-arrival time is exponential. Service time is also exponential.

a) What is probability that next arrival is before 1 PM?

$\lambda = 1$, $t = 1$, So, $\lambda t = 1$. reqd. prob. = $P(x \le t) = 1 - e^{-\lambda t} = 1 - e^{-1} = 0.632$

b) What is probability that next arrival is between 1 and 2 PM?

= Prob of arrival before 2 PM – Prob of arrival before 1 PM

$= (1 - e^{-2}) - (1 - e^{-1}) = (1 - 0.135) - (1 - 0.368) = 0.233$

c) What is probability that next arrival after 2 PM?

Reqd. Prob. = $P(X > 2) = e^{-2} = 0.135$

d) Nobody arrived till 1 PM. Probability of next arrival before 2 PM?

Using memory-less property, Reqd. Prob. = $1 - e^{-1} = 0.632$

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Queuing Example

In a parking lot on a Sunday, last arrival was at 12 noon. Arrival rate λ = 1/hour; service rate μ = 2/hour. No. of arrivals is Poisson and Inter-arrival time is exponential. Service time is also exponential.

e) Server is serving at 1 PM. Prob. that it is not complete by 2 PM?

μ = 2/hour. Using memory-less property, t = 1 hour
Reqd. Prob. = $e^{-2*1} = e^{-2} = 0.135$

f) Server is serving at 1 PM. Prob. that it is not complete by 1.10 PM?

μ = 2/hour. Using memory-less property, t = 10/60 = 1/6 hour
Reqd. Prob. = $e^{-2*1/6} = e^{-1/3} = 0.717$

g) Server is serving at 1 PM. Prob. that it will be complete by 1.10 PM?

μ = 2/hour. Using memory-less property, t = 10/60 = 1/6 hour
Reqd. Prob. = $(1 - e^{-2*1/6}) = (1 - e^{-1/3}) = 1 - 0.717 = 0.283$

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Queuing Example

In a parking lot on a Sunday, last arrival was at 12 noon. Arrival rate λ = 1/hour; service rate μ = 2/hour. No. of arrivals is Poisson and Inter-arrival time is exponential. Service time is also exponential.

e) Server is serving at 1 PM. Prob. that it is not complete by 2 PM?

f) Server is serving at 1 PM. Prob. that it is not complete by 1.10 PM?

g) Server is serving at 1 PM. Prob. that it will be complete by 1.10 PM?

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M/D/1 and M/M/s Queuing Models

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M/D/1 Model

In this model, the Arrival rate is Poisson but the service rate is deterministic – i.e. there is constant service rate.

Steady-State Parameters can be obtained by remembering that Lq is one half of that of M/M/1 system.

We know for M/M/1 system, $L_q = \rho^2/(1-\rho) = L - \rho$

Hence, the following formulas are written for the M/D/1 Model:

$$L_q = (1/2)*\rho^2/(1-\rho)$$

$$L = L_q + \rho = \rho + (1/2)*\rho^2/(1-\rho)$$

Rest of the parameters may be obtained from Little's Formula,

$$W = L / \lambda$$

$$W_q = L_q / \lambda$$

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M/M/1 vs M/D/1 Example

Arrivals to an airport with a single runway are poisson distributed with a rate of 30 per hour. The average time to land an aircraft is 90 seconds and this time is exponentially distributed.

a) Find the utilization of the runway and the steady state parameters L (Length of system) and W (Average waiting time in the system).

b) Will your answer change if time to land an aircraft is a Constant equal to 90 seconds? By how much?

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M/M/1 Consideration

Arrivals to an airport with a single runway are poisson distributed with a rate of 30 per hour. The average time to land an aircraft is 90 seconds and this time is exponentially distributed. Find the utilization of the runway and the steady state parameters L (Length of system) and W (Average waiting time in the system).

Answer:

Arrival rate $\lambda = 0.5$ per minute (Arrival is 30 per minute)

Service rate $\mu = 60/90$ per minute = 2/3 per minute.

Runway utilization (with a single server M/M/1 queue)

$\rho = \lambda / \mu = (1/2)/(2/3) = 3/4 = 75\ \%$

Steady State Parameters:

Length of system $L = \rho / (1 - \rho) = (3/4)/(1 - 3/4) = (3/4)/(1/4) = 3$ aircrafts.

Average waiting time in system: $W = L / \lambda = 3/(1/2) = 6$ minutes.

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M/D/1 Consideration

Arrivals to an airport with a single runway are poisson distributed with a rate of 30 per hour. The time to land an aircraft is a constant 90 seconds. Find the utilization of the runway and the steady state parameters L (Length of system) and W (Average waiting time in the system).

Answer:

Arrival rate λ = 0.5 per minute (Arrival is 30 per minute)

Service rate μ = 60/90 per minute = 2/3 per minute.

Runway utilization (single server M/D/1 queue)

- **$\rho = \lambda / \mu = (1/2)/(2/3) = 3/4 = 75\%$ - same as the previous value.**

Steady-state parameters will change as this is a M/D/1 queue.

Length of System $L = \rho + (1/2)^* \rho^2/(1 - \rho) = (3/4) + (1/2)^*(3/4)^2/(1 - 3/4)$
$= (3/4) + (1/2)^*(9/16)/(1/4) = (3/4) + (9/8) = 15/8 = 1.875$ aircrafts.

Average waiting time in system: $W = L/\lambda = (15/8)/(1/2)=15/4 = 3.75$ mins

Note: L reduced from 3 to 1.875 and W reduced from 6 to 3.75 mins.

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Another M/M/1 vs M/D/1

Cars arrive at a service station at an average rate of 1 per three-hour period. The average time the mechanics require to fix a car is about one and a half hour. If the arrival is Poisson and the service time is exponentially distributed,

a) Find the utilization of the service station and the average waiting time of a car in the service station.

b) Will your answer change if time to fix a car is a Constant equal to one and a half hour? By how much?

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M/M/1 Consideration

Cars arrive at a service station at an average rate of 1 per three-hour period. The average time the mechanics require to fix a car is about one and a half hour. If the arrival is Poisson and the service time is exponentially distributed, find the utilization of the service station and the average waiting time of a car in the service station.

Answer:

In this problem, we have a single server M/M/1 queue.

Arrival rate of cars $\lambda = 1/3$ per hour (Arrival is 1 per 3-hour period)

Service rate $\mu = 1/1.5$ per hour = 2/3 per hour.

Utilization of the service station:

$\rho = \lambda / \mu = (1/3)/(2/3) = 1/2 = 50\ \%$

Length of the system $L = \rho / (1 - \rho) = (1/2)/(1 - 1/2) = (1/2)/(1/2) = 1$ car.

Average waiting time of a car in the system will be:

$W = L / \lambda = 1/(1/3) = 3$ hours.

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M/D/1 Consideration

Cars arrive at a service station at an average rate of 1 per 3-hour period. The time the mechanics require to fix a car is a constant one and a half hour. If the arrival is poisson, find service station utilization and average waiting time of a car in service station.

Answer:

Arrival rate of cars $\lambda = 1/3$ per hour (as arrival is 1 per 3-hour period)

Service rate $\mu = 1/1.5$ per hour = 2/3 per hour.

Utilization of the service station (single server M/D/1 queue)

- **$\rho = \lambda/\mu = (1/3)/(2/3) = 1/2 = 50\ \%$ - same as the previous value.**

Steady-state parameters will change as this is a M/D/1 queue.

Length of System $L = \rho + (1/2)^* \rho^2/(1 - \rho) = (1/2) + (1/2)^*(1/2)^2/(1 - 1/2)$
$= (1/2) + (1/2)^*(1/4)/(1/2) = (1/2) + (1/4) = 3/4 = 0.75$ car.

Average waiting time in the system: $W = L/\lambda = (3/4)/(1/3) = 9/4$ hours

Note: L reduced from 1 car to 0.75 car per hour and W reduced from 3 hours to 2 hours 15 minutes!

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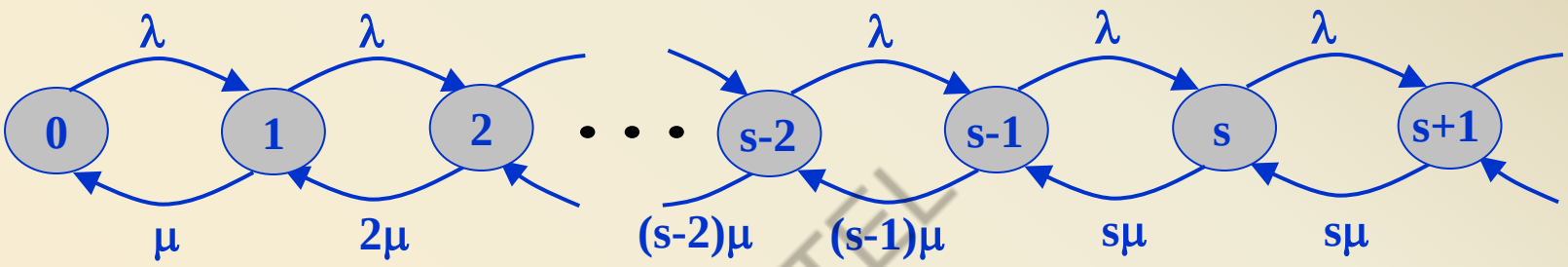
M/M/s Model

- **In this model, no of independent servers are s. The servers are all identical**



$$P_0 = \left(\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \cdot \frac{1}{1-(\lambda/(s\mu)} \right)^{-1}$$

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} P_0 & for \quad n = 1,2,\ldots,s \\ \dfrac{(\lambda/\mu)^n}{s!c^{n-s}} P_0 & for \quad n = s+1, s+2, \ldots \end{cases}$$

Steady State Condition

$\rho = (\lambda/s\mu) < 1$

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M/M/s Model

Steady state Condition

Utilization factor $\rho = \lambda/(s\mu) < 1$

Expected No. of Customers in Queue (Lq)

$$L_q = \sum_{n=s}^{\infty}(n-s)P_n = \ldots = \frac{(\lambda/\mu)^s \rho}{s!(1-\rho)^2}P_0$$

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Little's Formula

Expected Waiting time is queue: $W_q = L_q/\lambda$

Expected Waiting time in system: $W = W_q + (1/\mu)$

Expected No, of Customers in system: $L = \lambda W$

$$L = \lambda W = \lambda(W_q + 1/\mu) = L_q + \lambda/\mu$$

M/M/2 Example

A small bank has two counters – one for deposit and another for withdrawal of money. Customer arrival is Poisson at 10 per hour to the deposit and 20 per hour to the withdrawal counter.

Exponentially distributed service time in each counter is 2 minutes per customer.

Find average waiting time of a customer in the system for deposit counter as well as for withdrawal counter with and without pooling.

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M/M/2 Example

A small bank has two counters – one for deposit and another for withdrawal of money. Customer arrival is Poisson at 10 per hour to the deposit and 20 per hour to the withdrawal counter. Exponentially distributed service time in each counter is 2 minutes per customer. Find average waiting time of a customer in the system for deposit counter as well as for withdrawal counter with and without pooling.

Answer: **Without pooling of resources**

Deposit Counter:

Arrival rate λ = 10/hour; Service rate = 2 mins per customer = 30/hour
System utilization factor, $\rho = \lambda/\mu$ = 10/30 = 1/3
Average no. of customers in system, $L = \rho/(1 - \rho)$ = (1/3)/(1 – 1/3) = 1/2
Average waiting time in system $W = L/\lambda$ = (1/2)/10 = 1/20 Hour = 3 mins

Withdrawal Counter:

Arrival rate λ = 20/hour; Service rate = 2 mins per customer = 30/hour
System utilization factor, $\rho = \lambda/\mu$ = 20/30 = 2/3
Average no. of customers in system, $L = \rho/(1 - \rho)$ = (2/3)/(1 – 2/3) = 2
Average waiting time in system $W = L/\lambda$ = 2/20 = 1/10 Hour = 6 mins

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M/M/2 Example

With pooling of resources

Arrival rate of customers at the counters, λ = 10+20 = 30 per hour
Service rate at any one counter = 2 minutes = 30 per hour

Hence, we have, system utilization factor, ρ = λ/sμ = 30/(2*30) = 1/2

Using the M/M/2 queuing formula, we have,

$$P_0 = \left(\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \cdot \frac{1}{1-(\lambda/s\mu)} \right)^{-1}$$

$=[((30/30)^0/0!)+(30/30)^1/1!)+(30/30)^2/2!)*(1/(1-1/2))]^{-1} = [1+1+(1/2)*2]^{-1} = 1/3$

$$L_q = \sum_{n=s}^{\infty} (n-s)P_n = \ldots = \frac{(\lambda/\mu)^s \rho}{s!(1-\rho)^2} P_0 = \frac{(30/30)^2 * (1/2)}{2!(1-1/2)^2} * (1/3) = \frac{(1/2)}{2*1/4} * (1/3) = \frac{1}{3}$$

Wq = Lq/λ =(1/3)/30 =1/90;

W = Wq+(1/μ) =(1/90)+(1/30)=2/45 = 2.67 mins.

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Pooling of Resources

With pooling of resources, there is an improvement in the performance of both the counters.

In the withdrawal counter, the waiting time has reduced from 6 minutes to 2.67 minutes per customer.

While at the deposit counter, the waiting time has reduced from 3 minutes to 2.67 minutes per customer.

Advantage of Pooling of Resources

Pooling of resources always creates improvements in the system performance.

This is because when the two counters are not pooled, one counter may be idle while the other is busy. The work goes on at a rate equivalent to only one counter.

With pooling, when there are no deposit customers, both counters can carry out withdrawal work – thus increasing efficiency.

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M/M/s and M/M/∞ Models

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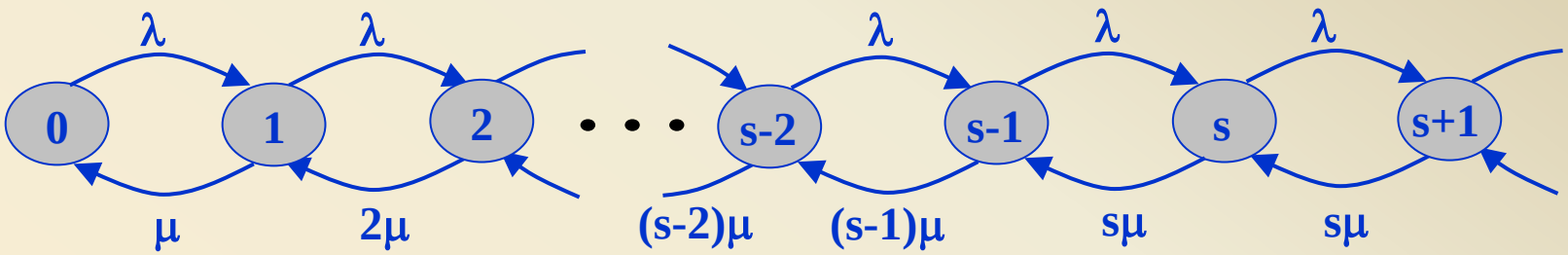
M/M/s Model

- **No. of independent and identical servers are s.**



$$P_0 = \left(\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \cdot \frac{1}{1-(\lambda/(s\mu)} \right)^{-1}$$

$$P_n = \begin{cases} \dfrac{(\lambda/\mu)^n}{n!} P_0 & for \quad n = 1,2,\ldots,s \\ \dfrac{(\lambda/\mu)^n}{s!\,s^{n-s}} P_0 & for \quad n = s+1, s+2, \ldots \end{cases}$$

$$L_q = \sum_{n=s}^{\infty} (n-s) P_n = \ldots = \frac{(\lambda/\mu)^s \rho}{s!(1-\rho)^2} P_0$$

Utilization factor

$\rho = (\lambda/s\mu) < 1$

Little's Formula

$W_q = L_q/\lambda$

$W = W_q + (1/\mu)$

$L = \lambda W = \lambda(W_q + 1/\mu)$

$= L_q + \lambda/\mu$

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M/M/1 vs M/M/2 Example

A small bank has two counters – one for deposit and another for withdrawal of money. Customer arrival is Poisson at 10 per hour to the deposit and 20 per hour to the withdrawal counter. Exponentially distributed service time in each counter is 2 minutes per customer. Find average waiting time of a customer in the system for deposit counter as well as for withdrawal counter with and without pooling.

Answer: **Without pooling of resources**

Deposit Counter:

Arrival rate λ = 10/hour; Service rate = 2 mins per customer = 30/hour
System utilization factor, $\rho = \lambda/\mu$ = 10/30 = 1/3
Average no. of customers in system, $L = \rho/(1 - \rho)$ = (1/3)/(1 – 1/3) = 1/2
Average waiting time in system $W = L/\lambda$ = (1/2)/10 = 1/20 Hour = 3 mins

Withdrawal Counter:

Arrival rate λ = 20/hour; Service rate = 2 mins per customer = 30/hour
System utilization factor, $\rho = \lambda/\mu$ = 20/30 = 2/3
Average no. of customers in system, $L = \rho/(1 - \rho)$ = (2/3)/(1 – 2/3) = 2
Average waiting time in system $W = L/\lambda$ = 2/20 = 1/10 Hour = 6 mins

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M/M/1 vs M/M/2 Example

With pooling of resources

Arrival rate of customers at the counters, λ = 10+20 = 30 per hour
Service rate at any one counter = 2 minutes = 30 per hour

Hence, we have, system utilization factor, ρ = λ/sμ = 30/(2*30) = 1/2

Using the M/M/2 queuing formula, we have,

$$P_0 = \left(\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \cdot \frac{1}{1-(\lambda/s\mu)} \right)^{-1}$$

$=[((30/30)^0/0!)+(30/30)^1/1!)+(30/30)^2/2!)*(1/(1-1/2))]^{-1} = [1+1+(1/2)*2]^{-1} = 1/3$

$$L_q = \sum_{n=s}^{\infty} (n-s)P_n = \ldots = \frac{(\lambda/\mu)^s \rho}{s!(1-\rho)^2} P_0 = \frac{(30/30)^2 * (1/2)}{2!(1-1/2)^2} * (1/3) = \frac{(1/2)}{2*1/4} * (1/3) = \frac{1}{3}$$

Wq = Lq/λ =(1/3)/30 =1/90;

W = Wq+(1/μ) =(1/90)+(1/30)=2/45 = 2.667 mins.

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Pooling of Resources

Advantage of Pooling of Resources

Pooling of resources always creates improvements in the system performance.

This is because when the two counters are not pooled, one counter may be idle while the other is busy. The work goes on at a rate equivalent to only one counter.

With pooling, when there are no deposit customers, both counters can carry out withdrawal work – thus increasing efficiency.

Pooling of Resources help in achieving:

- **Balanced utilization**
- **Shorter waiting time**
- **Ease of customer operation**

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M/M/∞ or Self-Service Model

In the M/M/∞ model, number of identical servers are infinity. This reduces to the self-service model.

The steady state system parameters are as given below.

Let $a = \frac{\lambda}{\mu}$

$$P_0 = e^{-a} = e^{-\frac{\lambda}{\mu}}$$

$$P_n = \frac{e^{-a} a^n}{n!}$$ **where n = 1, 2, 3 etc.**

$$L = a = \frac{\lambda}{\mu}$$

$$L_q = 0$$

$$W = \frac{L}{\lambda} = \frac{1}{\mu}$$

$$W_q = \frac{L_q}{\lambda} = 0$$

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M/M/∞ or Self-Service Model

In a small bank, customer arrival is Poisson at 30 per hour. In this back, the customers do self-service. Exponentially distributed service time 2 minutes per customer. Find the extent to which the bank is busy with customers. Also find the average number of customers and the average waiting time of a customer in the bank.

Answer: **Self-Service System**

Arrival rate λ = 30/hour; Service rate = 2 mins per customer = 30/hour
Hence, λ/μ = 30/30 = 1

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So, $P_0 = e^{-\frac{\lambda}{\mu}} = e^{-1} = 0.368$ **Hence, Busy Period = 1–0.368 = 0.632 = 63.2%**

Average No. of customers in the bank: $L = a = \frac{\lambda}{\mu} = 1$

Average waiting time in the bank: $W = \frac{L}{\lambda} = \frac{1}{\mu} = \frac{1}{30}$ **hour = 2 minutes.**

Comparison of M/M/1, M/M/2, and M/M/∞

Parameter	M/M/1 Deposit	M/M/1 Withdrawal	M/M/2	M/M/∞ Self-Service
Arrival Rate	λ = 10/hour	λ = 20/hour	λ = 30/hour	λ = 30/hour
Service Rate	μ = 30/hour	μ = 30/hour	μ = 30/hour	μ = 30/hour
P_0	0.667	0.333	0.333	0.368
Busy Period	0.333	0.667	0.667	0.632
L	0.5	2	1.333	1
W	3 minutes	6 minutes	2.667 minutes	2 minutes

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Another Example

In a small city, there are two identical cab companies.

For each, Arrival rate of customers is 4 per hour and Average Service Time is 12 minutes.

Assuming Poisson arrival and exponential service time distribution, find out whether there will be change in the utilization factor ρ or Waiting time in queue Wq, if the two companies merge.

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Another Example

In a small city, there are two identical cab companies. For each, Arrival rate of customers is 4 per hour and Average Service Time is 12 minutes. Assuming Poisson arrival and exponential service time distribution, find out whether there will be change in the utilization factor ρ or Waiting time in queue Wq, if the two companies merge.

Answer: **Without pooling of resources at any one company**

For each company, Average arrival rate $\lambda = 4$ per hour.

Average service time $1/\mu = 12$ minutes, and hence $\mu = 5$ per hour.

Hence, the utilization factor, $\rho = \lambda/\mu = 4/5 = 0.80$

Average number of customers in the system,

$Lq = \rho^2/(1 - \rho) = (4/5)^2/(1 - 4/5) = 16/5$

Average waiting time in the queue for the customers,

$Wq = Lq/\lambda = (16/5)/4 = 16/20$ hour $= 4/5$ hour $= 48$ minutes.

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Another Example

With merging of the companies

Arrival rate of customers at the new company, λ = 4+4 = 8 per hour

There are 2 servers – Service rate at any one server = 5 per hour

Hence, system utilization factor, ρ = λ/sμ = 8/(2*5) = 8/10 = 4/5 = 0.80

Using the M/M/2 queuing formula, we have,

$$P_0 = \left(\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \cdot \frac{1}{1-(\lambda/s\mu)} \right)^{-1}$$

$P_0 = [((8/5)^0/0!) + (8/5)^1/1!) + (8/5)^2/2!)*(1/(1–4/5))]^{-1}$

$= [1 + (8/5) + (64/50)*5]^{-1} = [1 + (8/5) + (32/5)]^{-1} = 1/9$

$$L_q = \sum_{n=s}^{\infty} (n-s)P_n = \ldots = \frac{(\lambda/\mu)^s \rho}{s!(1-\rho)^2} P_0 = \frac{(8/5)^2 * (4/5)}{2!(1-4/5)^2} * (1/9) = \frac{256/125}{2/25} * (1/9) = \frac{128}{45}$$

Hence, Wq = Lq/λ = 128/(45*8) = 128/360 = 16/45 hour = 21.33 minutes

Note: With merging of companies, Utilization remains same at 0.8 but Wq reduces from 48 minutes to 21.33 minutes!

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Self-Service Model

In a small city, a company gives cabs for customers to ride on their own for a fee. There is no constraint on number of vehicles available. Arrival rate of customers is 8 per hour and Average Service Time is 12 minutes. Assuming Poisson arrival and exponential service time distribution, find out the busy period and waiting time in queue Wq.

Answer: **Self-Service System**

Arrival rate λ = 8/hour; Service rate = 12 mins per customer = 5/hour
Hence, λ/μ = 8/5 = 1.6

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So, $P_0 = e^{-\frac{\lambda}{\mu}} = e^{-1.6} = 0.202$

Hence, Busy Period = 1–0.202 = 0.798 = 79.8%

Average waiting time in queue: $W_q = 0$

Comparison of M/M/1, M/M/2, and M/M/∞

Parameter	M/M/1 Each Company	M/M/2 Pooled Company	M/M/∞ Self-Service
Arrival Rate	λ = 4/hour	λ = 8/hour	λ = 8/hour
Service Rate	μ = 5/hour	μ = 5/hour	μ = 5/hour
P_0	0.2	1/9 = 0.111	0.202
Busy Period	0.8	0.889	0.798
W_q	48 minutes	21.333 minutes	0 minutes

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Finite Queue Space Model

A service station has only one pump. Cars wanting gas arrive as per Poisson process at 20/hour. Only 4 cars can be in the pump including the queue.

If there are n cars in the station, probability that an arriving car will balk is n/4 for n = 1, 2, 3, 4.

Time to fill a car is exponential with a mean of 3 minutes.

a) Construct the rate diagram

b) Develop equations

c) Find the expected waiting times for the cars that stay.

NPTEL

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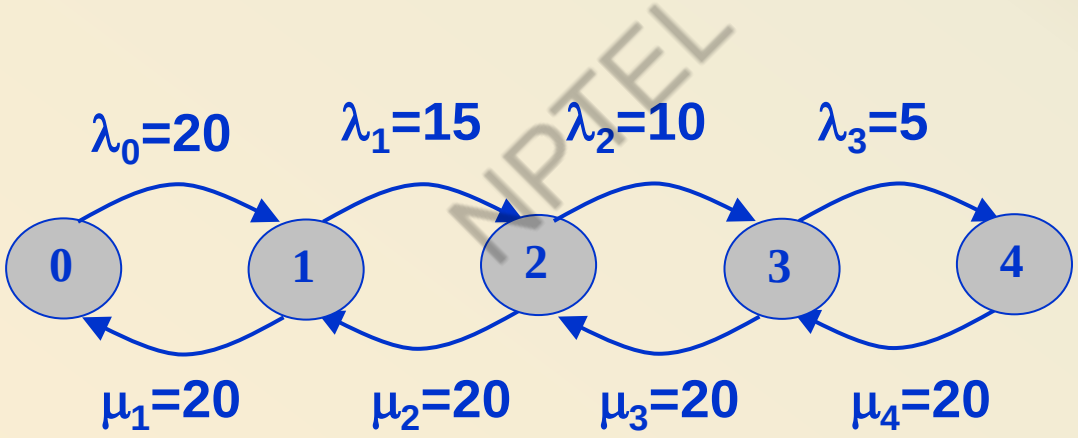
Answer: Rate Diagram



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Why are λ values reducing?

This is the effect of Balking. It is given that with n cars in the station, probability that an arriving car will balk is n/4

Finite Queue Space and Queuing Cost Models

NPTEL

Finite Queue Space Model

A service station has only one pump. Cars wanting gas arrive as per Poisson process at 20/hour. Only 4 cars can be in the pump including the queue.

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NPTEL

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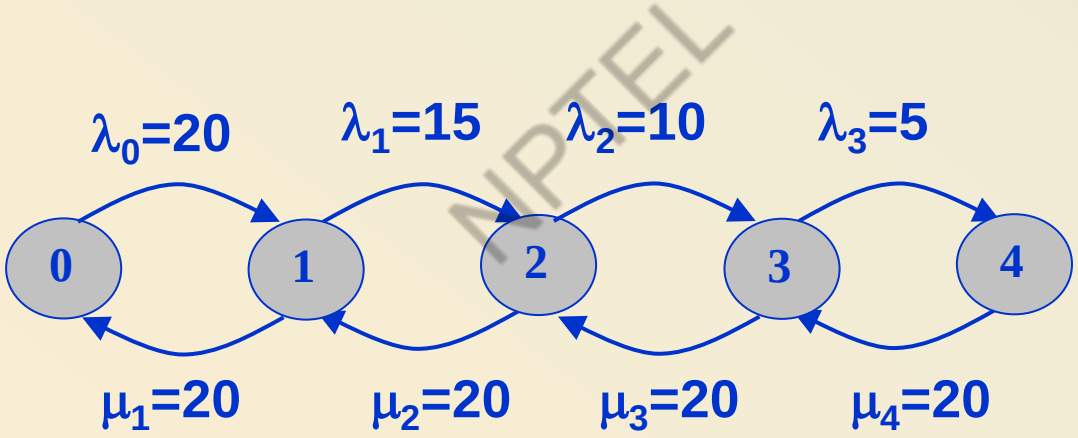

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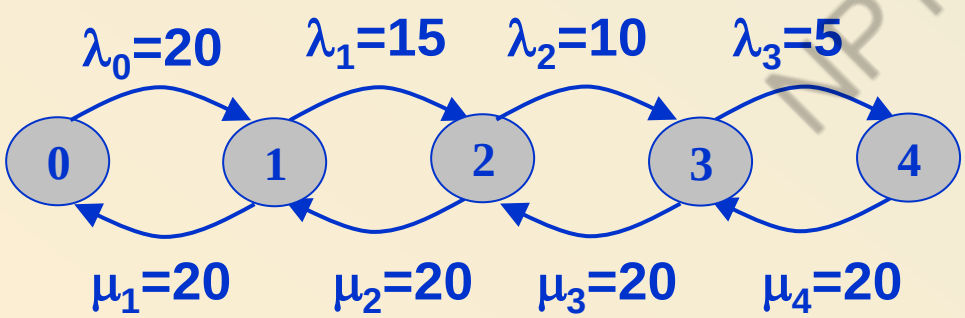


Equations (In=Out)

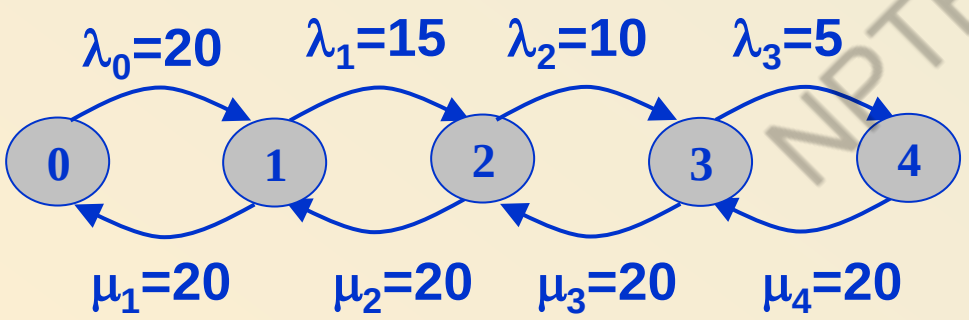
At Node 0: $20P_1 = 20\,P_0$
At Node 1: $20P_0+20P_2 = 35P_1$
At Node 2: $15P_1+20P_3 = 30P_2$
At Node 3: $10P_2+20P_4 = 25P_3$
At Node 4: $5P_3 = 20P_4$
Also: $P_0+P_1+P_2+P_3+P_4 = 1$

NPTEL

Finite Queue Space Model

A service station has only one pump. Cars wanting gas arrive as per Poisson process at 20/hour. Only 4 cars can be in the pump including the queue. If there are n cars in the station, probability that an arriving car will balk is n/4 for n = 1, 2, 3, 4. Time to fill a car is exponential with a mean of 3 minutes. a) Construct the rate diagram. b) Develop equations, c) Find the expected waiting times for the cars that stay.

Answer: Rate Diagram



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At Node 4: $5P_3 = 20P_4$
Also: $P_0+P_1+P_2+P_3+P_4 = 1$

Solving the equations:

$P_1=P_0$; $20P_2=15P_1$ i.e. $P_2=(3/4)P_1=(3/4)P_0$

$20P_3=(30–20)P_2$ i.e. $P_3=(1/2)P_2=(3/8)P_0$; $20P_4=5P_3$ i.e. $P_4=(1/4)P_3=(3/32)P_0$

$P_0+P_1+P_2+P_3+P_4=1$ So, $(1+1+(3/4)+(3/8)+(3/32))\,P_0=1$; so $P_0=32/103$

Hence, $P_0 = 32/103$; $P_1 = 32/102$; $P_2 = 24/103$; $P_3 = 12/103$; $P_4 = 3/103$

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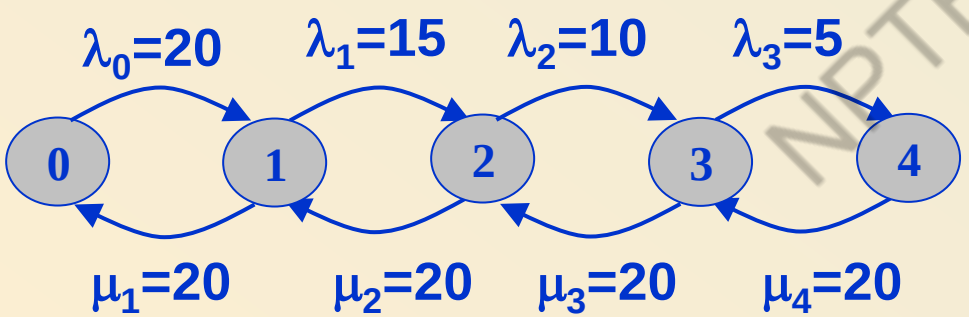
Finite Queue Space Model

A service station has only one pump. Cars wanting gas arrive as per Poisson process at 20/hour. Only 4 cars can be in the pump including the queue. If there are n cars in the station, probability that an arriving car will balk is n/4 for n = 1, 2, 3, 4. Time to fill a car is exponential with a mean of 3 minutes. a) Construct the rate diagram. b) Develop equations, c) Find the expected waiting times for the cars that stay.

Answer: Rate Diagram



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At Node 4: $5P_3 = 20P_4$

Also: $P_0+P_1+P_2+P_3+P_4 = 1$

Solving the equations:

$P_0 = 32/103$; $P_1 = 32/102$; $P_2 = 24/103$; $P_3 = 12/103$; $P_4 = 3/103$

So, $L = \sum_{0}^{4} nP_n = 0P_0+1P_1+2P_2+3P_3+4P_4 = 128/103$

$\bar{\lambda} = \sum_{n=0}^{4} \lambda_n P_n = 20P_0+15P_1+10P_2+5P_3+0P_4 = 1420/103$

Hence $W = \frac{L}{\bar{\lambda}}$

= 128/1420

= 0.090 hours

NPTEL

Finite Queue Space Model

Two agents book airline reservations using 2 phones. In addition, a called can be put on hold.

If all three lines are busy, customers balk.

Poisson arrival is 1 per min, Exponential service time is 0.5 min.

Find probabilities for:

a) a caller gets to talk immediately,

b) caller will be put on hold,

c) caller gets a busy signal.

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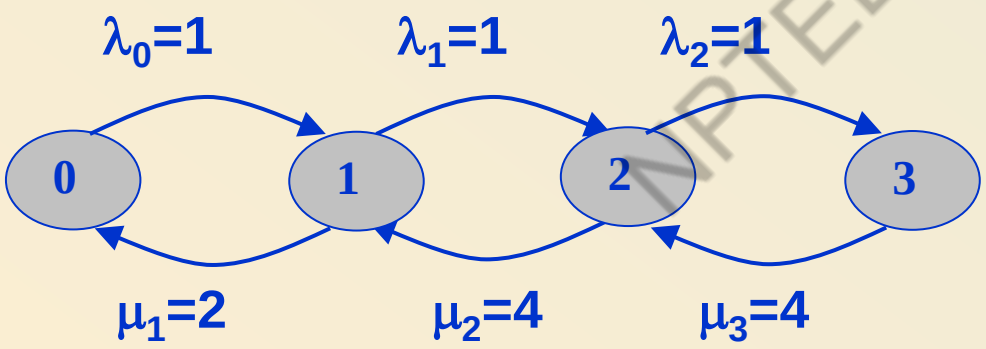
Another Finite Queue Space Model

Two agents book airline reservations using 2 phones. In addition, a called can be put on hold. If all three lines are busy, customers balk. Poisson arrival is 1 per min, Exponential service time is 0.5 min. Find probabilities for: a) a caller gets to talk immediately, b) caller will be put on hold, c) caller gets a busy signal.

The Rate Diagram



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Why no arrival beyond n=3? **Because customers balk**

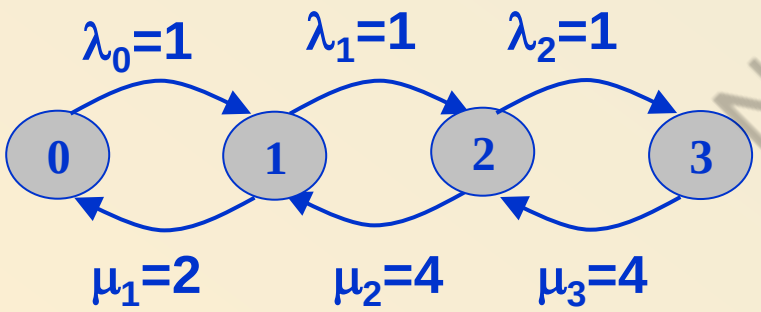
Why service is less for n=1? **Because there is only one customer**

Another Finite Queue Space Model

Two agents book airline reservations using 2 phones. In addition, a called can be put on hold. If all three lines are busy, customers balk. Poisson arrival is 1 per min, Exponential service time is 0.5 min. Find probabilities for: a) a caller gets to talk immediately, b) caller will be put on hold, c) caller gets a busy signal.

Answer: λ = 1/min; μ = 2/min for 1 customer; 4/min for more customers

Rate Diagram



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At Node 2: $1P_1+4P_3 = 5P_2$

At Node 3: $1P_2 = 4P_3$

Also: $P_0+P_1+P_2+P_3 = 1$

NPTEL

Another Finite Queue Space Model

Two agents book airline reservations using 2 phones. In addition, a called can be put on hold. If all three lines are busy, customers balk. Poisson arrival is 1 per min, Exponential service time is 0.5 min. Find probabilities for: a) a caller gets to talk immediately, b) caller will be put on hold, c) caller gets a busy signal.

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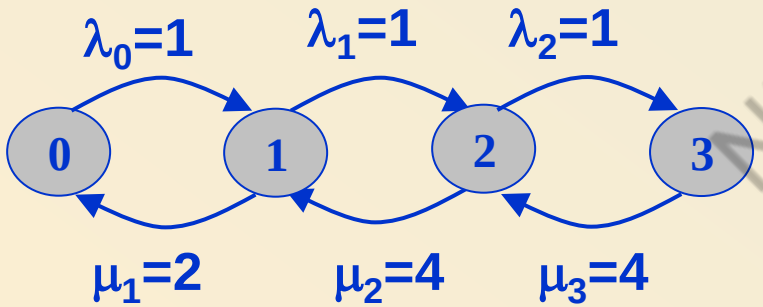
Rate Diagram



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At Node 3: $1P_2 = 4P_3$

Also: $P_0+P_1+P_2+P_3 = 1$

Solving the equations:

$2P_1 = 1P_0$ i.e. $P_1=(1/2)P_0$; $4P_2=3P_1-1P_0$ i.e. $P_2=((1/4)*(3/2)-1)P_0=(1/8)P_0$

$4P_3=1P_2$ i.e. $P_3=(1/4)P_2=(1/32)P_0$

$P_0+P_1+P_2+P_3=1$ So, $(1+(1/2)+(1/8)+(1/32))P_0=1$; so $P_0=32/53$

Hence, $P_0 = 32/53$; $P_1 = 16/53$; $P_2 = 4/53$; $P_3 = 1/53$

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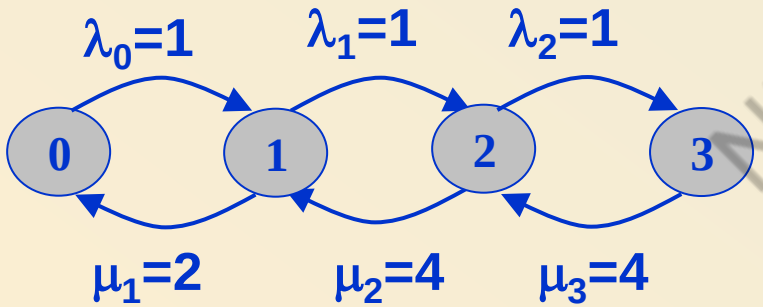
Another Finite Queue Space Model

Two agents book airline reservations using 2 phones. In addition, a called can be put on hold. If all three lines are busy, customers balk. Poisson arrival is 1 per min, Exponential service time is 0.5 min. Find probabilities for: a) a caller gets to talk immediately, b) caller will be put on hold, c) caller gets a busy signal.

Answer: λ = 1/min; μ = 2/min for 1 customer; 4/min for more customers

Rate Diagram



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Also: $P_0+P_1+P_2+P_3 = 1$

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Solving the equations: $P_0 = 32/53$; $P_1 = 16/53$; $P_2 = 4/53$; $P_3 = 1/53$

Hence, the probability values will be:

a) A caller gets to talk immediately: $P_0 + P_1 = (32/53)+(16/53) = 48/53$
b) Caller will be put to hold: $P_2 = 4/53$
c) Caller gets a busy signal: $P_3 = 1/53$

Another Finite Queue Space Model

A small car wash is to be opened where cars may arrive every 4 minutes on the average and the average service time is 3 minutes. Find the expected fraction of potential customers lost (EF), if the queue space is a) 0 and b) 2. Cars will go away if there is no space.

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Queuing Cost Model

Two types of costs are usually involved in a queuing model. These are:

1) Service Costs (SC)
2) Waiting Costs (WC)

A trade-off is necessary to strike a balance between the two costs and obtain optimum service capacity.

When the service facility is not adequate, the WC values go up. Augmentation of service facility is then required. This reduces WC but increases SC.

It is therefore important to obtain trade-off so as minimize Total Costs (TC) which is equal to SC+WC.

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Queuing Cost Model

Service Costs (SC)

In order to manage service costs, the consideration of the following are necessary:

- **Capital investment costs - interests, loan repayments**
- **Cost of operation - labor, energy, materials**
- **Cost of maintenance and repair**
- **Other costs - insurance, taxes, rental of the space**

Waiting Cost (WC)

- **Cost of poor service**
- **Opportunity loss of customers leaving**
- **Loss of repeat business due to dissatisfaction**
- **Loss of customer relations (CRM)**

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Queuing Cost, Priority and Networking Models

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Queuing Cost Model

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Queuing Cost Model

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Queuing Cost Model

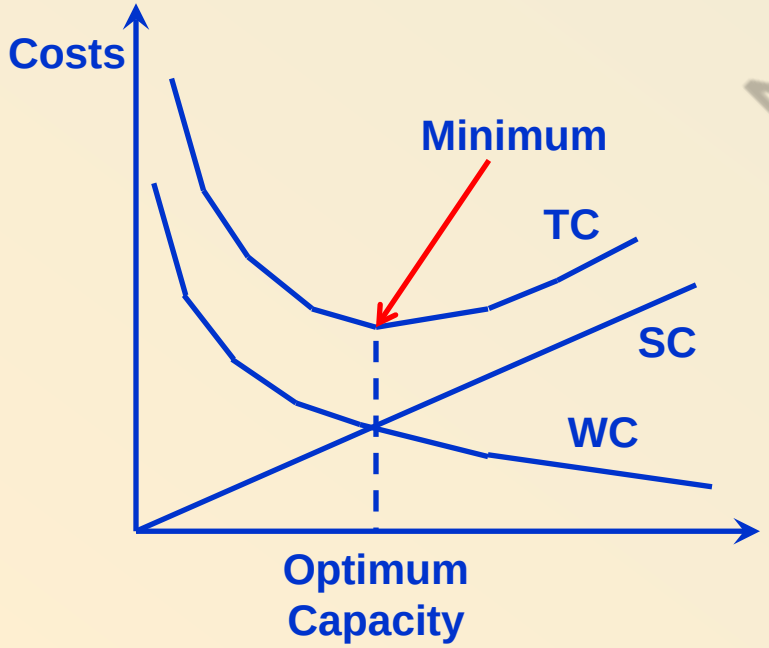
A simple analysis of queuing costs is presented below:

Total Queuing Cost TC = Service Cost (SC) + Waiting Cost (WC)

If cost of waiting of 1 customer for unit time is Cw

Then, Average waiting cost per customer = W*Cw

As λ customers arrive per unit time, WC = λ*W*Cw = L*Cw



Total cost (TC) will be minimum for optimum values of Service Cost (SC) and Waiting Cost (WC) and Optimum Capacity will be obtained.

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Queuing Cost Model Example

Service mechanics come to take spares at a shop at 6/hour on the average. Waiting for them costs Rs. 8/- per hour. A Shop attendant's wage is Rs. 5/- per hour. There is only one counter.

The service rates are: 1 attendant: 8/hour, 2 attendants: 12/hour, 3 attendants: 16/hour.

With usual M/M/1 assumptions, find how many attendants to choose.

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Queuing Cost Model Example

Service mechanics come to take spares at a shop at 6/hour on the average. Waiting for them costs Rs. 8/- per hour. A Shop attendant's wage is Rs. 5/- per hour. There is only one counter. The service rates are: 1 attendant: 8/hour, 2 attendants: 12/hour, 3 attendants: 16/hour. With usual M/M/1 assumptions, find how many attendants to choose.

Answer: First Compute ρ and W.

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No. of Attendants (N)	Arrival Rate per hour (λ)	Service Rate per hour (μ)	Utilization Factor ρ = λ/μ	Waiting time (W) in hour W = ρ/((1 – ρ)*λ)
1	6	8	6/8 = 3/4	(3/4)/(6*1/4) = 1/2
2	6	12	6/12 = 1/2	(1/2)/(6*1/2) = 1/6
3	6	16	6/16 = 3/8	(3/8)/(6*5/8) = 1/10

Queuing Cost Model Example

Service mechanics come to take spares at a shop at 6/hour on the average. Waiting for them costs Rs. 8/- per hour. A Shop attendant's wage is Rs. 5/- per hour. There is only one counter. The service rates are: 1 attendant: 8/hour, 2 attendants: 12/hour, 3 attendants: 16/hour. With usual M/M/1 assumptions, find how many attendants to choose.

Answer:

No. of Atten-dants (N)	Arrival Rate per hour (λ)	Service Rate per hour (μ)	Utilization Factor ρ = λ/ μ	Waiting time (W) in hour W = ρ/((1 – ρ)*λ)	Service Cost (Rs.) SC = N*8hr*5	Waiting Cost (Rs.) WC = W*8hr*λ*8	Total Cost TC = SC+WC
1	6	8	3/4	1/2	1*8*5 = 40	(1/2)*8*6*8 = 192	232
2	6	12	1/2	1/6	2*8*5 = 80	(1/6)*8*6*8 = 64	144 Lowest
3	6	16	3/8	1/10	3*8*5 = 120	(1/10)*8*6*8 = 38.4	158.4

Hence, We may choose 2 attendants – total cost is Rs. 144.

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Another Queuing Cost Model

Machines fail at 4 per hour and the cost of non-productive machine is Rs. 9 per hour.

A fast repairman charges Rs. 6 per hour and repairs at 7 per hour. A slow repairman charges Rs. 3 per hour but repairs at 5 per hour.

With usual M/M/1 assumptions, find which repairman to hire.

NPTEL

Another Queuing Cost Example

Machines fail at 4 per hour and the cost of non-productive machine is Rs. 9 per hour. A fast repairman charges Rs. 6 per hour and repairs at 7 per hour. A slow repairman charges Rs. 3 per hour but repairs at 5 per hour. With usual M/M/1 assumptions, find which repairman to hire.

Answer:

Repair-man	Arrival Rate per hour (λ)	Service Rate per hour (μ)	Utilization Factor $\rho = \lambda/\mu$	Average No. of machines in repair (L) $L = \rho/(1-\rho)$	M/c Idle Cost (Rs.) IC = L*8hr *hourly cost	Repairman Cost (Rs.) RC = 8 hr* hourly cost	Total Cost TC = SC+WC
Fast	4	7	4/7	(4/7)/(3/7) = 4/3	(4/3)*8*9 = 96	8*6 = 48	144 Lower
Slow	4	5	4/5	(4/5)/(1/5) = 4	4*8*9 = 288	8*3 = 24	312

Hence, We may choose the Fast Repairman – total cost is Rs. 144.

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Priority Discipline Queuing Models

- **Priority Discipline choices of a Queuing Model include: First Come First Served (FCFS), Service In Random Order (SIRO) and Last In First Served (LCFS) among others.**
- **Consider being served LCFS in a doctor's chamber! When it comes to people, FCFS is an automatic choice.**
- **However, think of stacking tyres and putting them onto vehicles – LCFS seems to be a natural choice!**
- **While choosing bolts and nuts for putting into assembly, SIRO seems to be most usual.**
- **Interesting to note that irrespective of the Priority Discipline choice, the expected values of steady state Performance measures remain the same.**

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Priority Discipline Queuing Models

- Average time (steady-state) that a customer spends in the system does not depend on the queue discipline:

$$E(W_{FCFS}) = E(W_{SIRO}) = E(W_{LCFS})$$

- However,

$$Var(W_{FCFS}) < Var(W_{SIRO}) < Var(W_{LCFS})$$

- Although average values are same, **variances** of waiting time **differ** for queuing models with FCFS, SIRO or LCFS.
- Relatively **large** waiting times are likely to occur with **LCFS** discipline and least likely to occur with FCFS discipline.

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Priority Discipline Queuing Models

- Models in which a customer's type determines the order in which customers undergo service are called priority queuing models.

- Priority Queuing Models are of two types:
- Nonpreemptive
- Preemptive

- In both these models, the order in which customers are served depends on the customer's "type".

- Hospital emergency rooms usually serve seriously ill patients before they serve nonemergency patients.

- Airport belts bring luggage of the priority customers before others.

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Priority Discipline Queuing Models

Nonpreemptive model

- a customer's service cannot be interrupted.
- After every service completion, the next customer to enter is chosen on the basis of a given priority for customer type.
- As per Kendall-Lee notation, steady state variables are L_k, L_{qk}, W_k *and* W_{qk} - where k stands for the type of customer.

Preemptive model

- A lower priority customer can be removed from service whenever a higher-priority customer arrives.
- Once no higher-priority customers are present, the removed customer reenters service.
- In a preemptive resume model, a customer's service continues from the point at which it was interrupted.
- In a preemptive repeat model, a customer begins service anew each time he or she reenters service.

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Finite Source Queuing Models

In Finite source or Machine repair models, arrivals to a system are drawn from a small population. The arrival rate will not be state-independent.

In the Machine repair problem, the system consists of *K* machines and *R* repair people.

A given machine may be in good or bad condition at an instant in time – the length of time that a machine remains in good condition follows an exponential distribution with rate λ.

Whenever a machine breaks down the machine is sent to a repair center consisting of *R* repair people.

The time it takes to complete repairs on a broken machine is assumed exponential with rate *μ.*

The calculations are quite involved and is beyond the scope.

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Queues in Series – Queuing Networks

- In a Queuing Network, a customer needs service in multiple stages – one after another. One such system is the k-stage series queuing system.
- For poisson arrival to a series queuing system and exponential service times and infinite queue space at each stage, it can be shown that interarrival times for each stage are exponential.
- It is required that enough capacities are available at each stage for above result to hold.
- For the entire system, L can be found by adding expected number of customers present at each stage. W can be found from Little's formula $L=\lambda W$.

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Conclusions

- **Variability in arrival as well as service leads to queues being formed although service rate is higher than arrival rate.**
- **When arrival rate is higher than service rate, no steady state is reached, and steady state analysis is not possible.**
- **In order to improve queuing model performance, it is necessary to identify and eliminate the variability in the process.**
- **Some measures are pooling of resources, resorting to self-service and so on.**

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Conclusions

- Variability leads to waiting times although utilization < 100%
- Customers balk when the queue size is finite, it is important to create space for queues to form.
- Queuing costs include cost of waiting and cost of service – a trade-off between the two is necessary in order to achieve optimal service capacity.
- Priority discipline considerations may be necessary to accommodate high-priority or emergency customers.

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THANK YOU

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